

GAP-ONLY INTERLACING OF THE REAL ROOTS OF CONSECUTIVE YABLONSKII–VOROB’EV POLYNOMIALS, GIVEN THE PUBLISHED REAL-ROOT COUNT

ABSTRACT. Let Y_n be the Yablonskii–Vorob’ev polynomials. In the Conclusions of their paper on the roots of generalised Okamoto polynomials, Roffelsen and Stokes record that the numbers of real roots of the Y_n are known and that interlacing of the real roots of Y_{n-1} and Y_{n+1} is a theorem of Clarkson, “but interlacing of roots of consecutive polynomials Y_n and Y_{n+1} has not been proved”. Their own definition of interlacing is a two-sided condition on gaps alone: between two real roots of one polynomial there lies a real root of the other. Granting three published facts about the Y_n — simplicity of the roots, coprimality of consecutive members, and Roffelsen’s count of the real roots — we prove the quoted statement for every $n \geq 1$ in that gap-only sense and in no other, and in the sharper form that the merged real-root list of Y_n and Y_{n+1} has exactly $n + 1$ elements, strictly alternates, and begins with a root of Y_{n+1} . The qualification is not idle. Under the stronger reading of interlacing that a reader may well have in mind, which also demands that the two root counts differ by one, that same merged word makes the statement false for every odd n , since $m_{n+1} = m_n$ there; under that reading nothing here settles the sentence quoted above, and we do not claim it does. A sign identity read off the Yablonskii–Vorob’ev recurrence forces the parity of the number of roots of Y_{n+1} in each gap between consecutive roots of Y_n , and the published count of real roots leaves no room for more than the forced minimum.

1. THE STATEMENT

The Yablonskii–Vorob’ev polynomials are defined by

$$(1) \quad Y_0 = 1, \quad Y_1 = z, \quad Y_{n+1}Y_{n-1} = zY_n^2 - 4(Y_nY_n'' - (Y_n')^2) \quad (n \geq 1),$$

and $Y_n \in \mathbb{Z}[z]$ is monic of degree $d_n = n(n+1)/2$. They carry the rational solutions of the second Painlevé equation. Write

$$\mathcal{Z}_k = \{x \in \mathbb{R} : Y_k(x) = 0\}, \quad m_k = \#\mathcal{Z}_k.$$

The Conclusions of Roffelsen and Stokes [5], quoted here from the e-print [arXiv:2402.15887v1](#), name one gap in the picture for the Yablonskii–Vorob’ev family:

the numbers of real roots of these polynomials Y_n are known
[4], as well as interlacing of roots of Y_{n-1} and Y_{n+1} on the real

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line [2], but interlacing of roots of consecutive polynomials Y_n and Y_{n+1} has not been proved.

The two citations inside the quotation are [4] and [2] of the present bibliography. The journal version of record was not consulted, so the wording above and the definition below are those of that e-print:

Definition 1 ([5]). The real roots of P and Q are *interlaced* on a connected $I \subseteq \mathbb{R}$ if between any two real roots of P in I there lies a real root of Q , and between any two real roots of Q in I there lies a real root of P .

This is a two-sided condition on gaps only; it does *not* require the two root counts to differ by one, and every assertion below is under it. Under a stronger variant that adds that demand, the word proved below makes the statement true for every even n and false for every odd n , since $m_{n+1} = m_n$ for odd n . So the sentence quoted above is settled here in the gap-only reading only; in the stronger reading it is not settled but refuted for odd n , and which of the two the quoted sentence intends we do not know.

Theorem 2. *For every $n \geq 1$ the real roots of Y_n and Y_{n+1} are interlaced on all of $\mathbb{R}$ in the sense of Definition 1. More precisely, $\mathcal{Z}_n \cup \mathcal{Z}_{n+1}$ has exactly $n+1$ elements, its elements alternate strictly between the two polynomials, and its least element belongs to $\mathcal{Z}_{n+1}$; for even n its greatest element also belongs to $\mathcal{Z}_{n+1}$. Equivalently, writing 1 for a root of Y_{n+1} and 0 for a root of Y_n and reading left to right along the real line, the merged root word is*

$$\underbrace{1010 \cdots 10}_{n+1} \quad (n \text{ odd}), \quad \underbrace{1010 \cdots 01}_{n+1} \quad (n \text{ even}).$$

2. THE INPUTS

Everything below uses only (1) and the following published facts, none of them ours:

- (F1) every root of Y_n is simple;
- (F2) $\gcd(Y_n, Y_{n+1}) = 1$;
- (F3) $m_k = \lfloor (k+1)/2 \rfloor$ for every $k \geq 0$;
- (F4) Y_k is monic of degree $d_k = k(k+1)/2$, hence $\text{sign } Y_k(+\infty) = +1$ and $\text{sign } Y_k(-\infty) = (-1)^{d_k}$.

(F1) and (F2) are due to Fukutani, Okamoto and Umemura [3]; see [1, Thm. 7.1] and [4, Thm. 4]. (F3) was conjectured by Clarkson [1, Conj. 7.4] and proved by Roffelsen [4, Thm. 1]. (F4) is immediate from (1).

3. THE PROOF

Lemma 3. *For every $n \geq 1$ and every $a \in \mathcal{Z}_n$,*

$$Y_{n+1}(a) Y_{n-1}(a) = 4 Y_n'(a)^2 > 0.$$

In particular Y_{n+1} and Y_{n-1} take the same, nonzero, sign at every real root of Y_n .

Proof. Rearranging (1),

$$(2) \quad Y_{n+1}Y_{n-1} - 4(Y'_n)^2 = Y_n(zY_n - 4Y''_n),$$

an identity in $\mathbb{Z}[z]$. The right-hand side vanishes at a , and $Y'_n(a) \neq 0$ because a is a simple root by (F1). $\square$

For an open interval I whose endpoints (possibly $\pm\infty$) are not roots of Y_k , write $N_k(I) = \#(\mathcal{Z}_k \cap I)$. Since all roots of Y_k are simple by (F1) and non-real roots occur in conjugate pairs, $N_k(I)$ is odd if and only if Y_k takes opposite signs at the two ends of I .

Lemma 4 (parity transfer). *Let $n \geq 1$ and $\mathcal{Z}_n = \{a_1 < \dots < a_{m_n}\}$ (nonempty, by (F3)). Put $L = (-\infty, a_1)$, $G_i = (a_i, a_{i+1})$ for $1 \leq i < m_n$, and $R = (a_{m_n}, +\infty)$. Then*

$$\begin{aligned} N_{n+1}(G_i) &\equiv N_{n-1}(G_i) \pmod{2} \quad (1 \leq i < m_n), \\ N_{n+1}(R) &\equiv N_{n-1}(R) \pmod{2}, \\ N_{n+1}(L) &\equiv N_{n-1}(L) + 1 \pmod{2}. \end{aligned}$$

Proof. By (F2), applied to the pairs (Y_{n-1}, Y_n) and (Y_n, Y_{n+1}) , no a_i is a root of Y_{n-1} or of Y_{n+1} , so all four counts are over intervals with non-root endpoints and the sign criterion above applies. By Lemma 3, Y_{n+1} and Y_{n-1} share their sign at each a_i ; by (F4) they are both positive at $+\infty$. Hence across each G_i , and across R , the two polynomials change sign together or not at all, which is the first two congruences. At $-\infty$, however, (F4) gives signs $(-1)^{d_{n+1}}$ and $(-1)^{d_{n-1}}$, and

$$d_{n+1} - d_{n-1} = \frac{(n+1)(n+2)}{2} - \frac{(n-1)n}{2} = 2n + 1$$

is odd, so those two signs are opposite while the signs at a_1 agree: exactly one of Y_{n+1}, Y_{n-1} changes sign across L . $\square$

Proof of Theorem 2. For $k \geq 0$ let $S(k)$ be the assertion

$\mathcal{Z}_k \cup \mathcal{Z}_{k+1}$ alternates strictly between $\mathcal{Z}_k$ and $\mathcal{Z}_{k+1}$, and its least element lies in $\mathcal{Z}_{k+1}$.

By (F3), $\#(\mathcal{Z}_k \cup \mathcal{Z}_{k+1}) = \lfloor \frac{k+1}{2} \rfloor + \lfloor \frac{k+2}{2} \rfloor = k+1$, so $S(k)$ determines the merged word completely: it is 1010... of length $k+1$, which is the word displayed in Theorem 2. We prove $S(n)$ for all $n \geq 0$ by induction.

$S(0)$ holds: $\mathcal{Z}_0 = \emptyset$ because $Y_0 = 1$, and $\mathcal{Z}_1 = \{0\}$.

Let $n \geq 1$ and assume $S(n-1)$. Keep the notation of Lemma 4. Reading $S(n-1)$ off, the merged word of $(\mathcal{Z}_{n-1}, \mathcal{Z}_n)$ has length n , alternates, and begins with an element of $\mathcal{Z}_n$; therefore

$$(3) \quad N_{n-1}(L) = 0, \quad N_{n-1}(G_i) = 1 \quad (1 \leq i < m_n), \quad N_{n-1}(R) = [n \text{ even}],$$

the last because the alternating word of length n starting in $\mathcal{Z}_n$ ends in $\mathcal{Z}_{n-1}$ exactly when n is even. Lemma 4 now transfers these to Y_{n+1} as parities,

hence as lower bounds:

$$(4) \quad N_{n+1}(L) \text{ odd}, \quad N_{n+1}(G_i) \text{ odd}, \quad N_{n+1}(R) \equiv [n \text{ even}] \pmod{2}.$$

Summing the forced minima over the $m_n + 1$ pieces of $\mathbb{R} \setminus \mathcal{Z}_n$,

$$(5) \quad \begin{aligned} m_{n+1} &= N_{n+1}(L) + \sum_i N_{n+1}(G_i) + N_{n+1}(R) \\ &\geq 1 + (m_n - 1) + [n \text{ even}] = m_n + [n \text{ even}]. \end{aligned}$$

But (F3) makes the right-hand side *equal* to m_{n+1} in both parities, because $\lfloor \frac{n+2}{2} \rfloor = \lfloor \frac{n+1}{2} \rfloor + [n \text{ even}]$. So (5) is an equality and every term is pinned at its minimum:

$$N_{n+1}(L) = 1, \quad N_{n+1}(G_i) = 1, \quad N_{n+1}(R) = [n \text{ even}].$$

For odd n the last value is forced to be 0 rather than merely at most 1 *by its parity* in (4). Thus $\mathcal{Z}_n \cup \mathcal{Z}_{n+1}$ reads

$$\bullet < a_1 < \bullet < a_2 < \cdots < a_{m_n} \quad (< \bullet \text{ iff } n \text{ even}),$$

with each $\bullet$ a single element of $\mathcal{Z}_{n+1}$: it alternates strictly and begins in $\mathcal{Z}_{n+1}$. That is $S(n)$.

Finally, $S(n)$ gives Definition 1 on $I = \mathbb{R}$: between any two real roots of Y_n there lie two *consecutive* ones, and alternation puts a root of Y_{n+1} between those; symmetrically with the roles exchanged. $\square$

4. RELATED STATEMENTS

(F3) is load-bearing above: parity alone bounds each interval count from below, and it is the published total that pins every one of them at its minimum. Our use of the index-2 pair is confined to Lemma 3, where only a sign is needed.

For the index-2 pair (Y_{n-1}, Y_{n+1}) an alternating chain is proved by Roffelsen [4, p. 6] and interlacing by Clarkson [1, Thm. 7.3], [2]; Roffelsen also gives $\min \mathcal{Z}_{n-1} > \min \mathcal{Z}_{n+1}$, $\max \mathcal{Z}_{n-1} < \max \mathcal{Z}_{n+1}$ [4, Thm. 2] and the separate counts of positive and negative real roots [4, Thm. 3]. Roffelsen and Stokes [5, Cor. 3.15] prove adjacent-direction interlacing for the generalised Okamoto polynomials $Q_{m,n}$, $Q_{m,n-1}$.

5. VERIFICATION

A program `verify.py` accompanies this note as a control on the interior of the induction, in exact integer and rational arithmetic and using only the Python standard library. It generates $Y_0, \dots, Y_{43}$ from (1) by exact division in $\mathbb{Z}[z]$ and checks the monicity and degrees (F4); the identity (2); (F1) and (F2) as squarefreeness and coprimality over finite fields; the real-root count (F3) by exact isolation with a Fujiwara bound for $n \leq 41$, where the expected value from (F3) is what certifies the isolation complete, with an independent Sturm computation that uses (F3) nowhere confirming the count only for $n \leq 12$; the occupancy statements (3) and their forced counterparts for Y_{n+1} ,

for $1 \leq n \leq 40$; and the arithmetic of (5) for $1 \leq n \leq 200000$. Its recorded run reports

VERDICT: ALL 24 CHECKS PASS

Three of those checks, the ones whose names begin **regression-**, compare the generated objects against coefficient lists, isolating intervals and merged root words held as literals inside the program itself; those literals are not displayed in this paper, and the checks are labelled in the run as the internal regression checks they are. No check asserts agreement with a display, a table or a numbered result that this paper does not contain. The program verifies no published proof and it verifies the root order only for $n \leq 40$; the theorem is the hand proof above and covers every n . It also confirms, for $1 \leq n \leq 200000$, the arithmetic behind the failure of the stronger reading of interlacing recorded after Definition 1.

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